

Mechatronics S4 A and S4B class notes –module 1 –day 2- Mechanical Engineering

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SOE CUSAT

SENSORS AND TRANSDUCERS

Module 1-continue

Sensor and transducer

- 'Sensor' is 'a device that detects a change in a physical stimulus and turns it into a signal which can be measured or recorded.
E.g. : Thermistor
- 'Transducer' is 'a device that transfers power from one system to another in the same or in the different form'.
E.g. Thermistor with it associate circuit convert heat to electricity.

As a comparison.....

'Sensor' for the sensing element itself and 'transducer' for the sensing element plus any associated circuitry. All transducers would thus contain a sensor and most (not all) sensors would also be transducers.

Sensing process

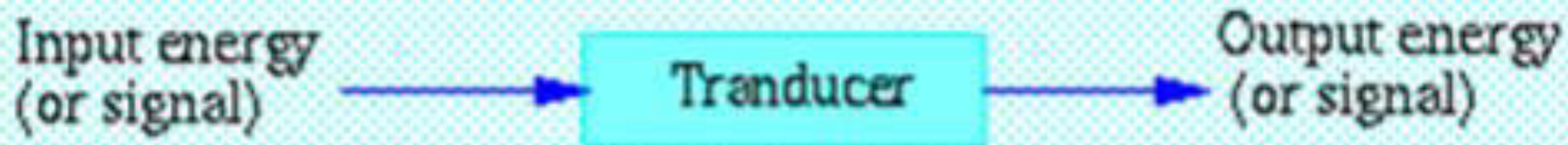


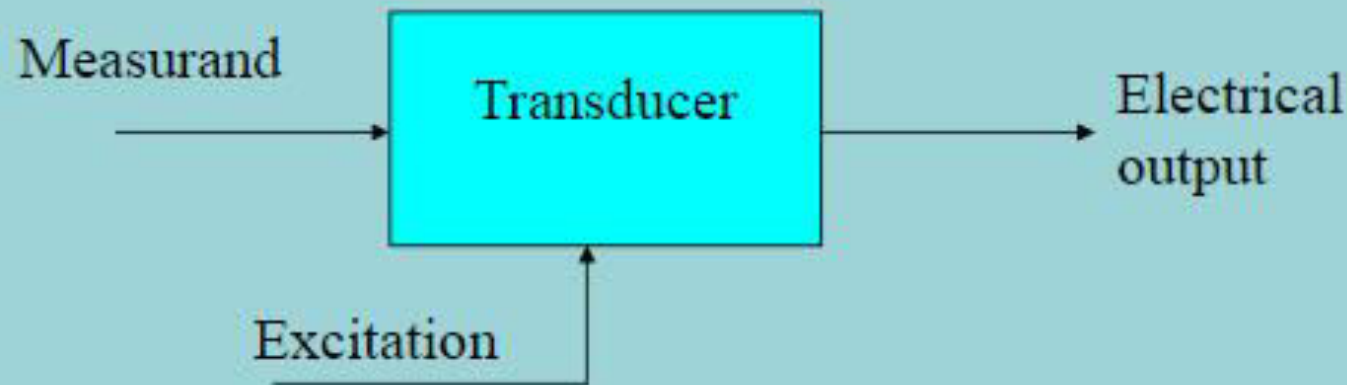
Figure 1 The sensing process

Definition of a transducer

Transducer is any device that converts energy in one form to another energy. The majority either convert **electrical energy to mechanical displacement** or convert some *non-electrical* physical quantity, such as temperature, sound or light to an electrical signal.

Functions of transducer

1. To *sense* the presence, magnitude, change in, and frequency of some *measurand*.
2. To *provide* an electrical output that, when appropriately processed and applied to readout device, gives accurate quantitative data about the *measurand*



Measurand – refers to the quantity, property or condition which the transducer translates to an electrical signal.

Classification of transducers

Transducer can be classified according to their application, based primarily on the physical quantity, property, or condition that is measured.

The transducer can be categories into:

A) Passive transducer :

- requires an external power
- output is a measure of some variation, such resistance and capacitance. E.g. : condenser microphone

B) Self generating transducer :

- not require an external power, and they produce analog voltage or current when stimulated by some physical form of energy. E.g. : Thermocouple

Specifications of a sensor/transducer system.

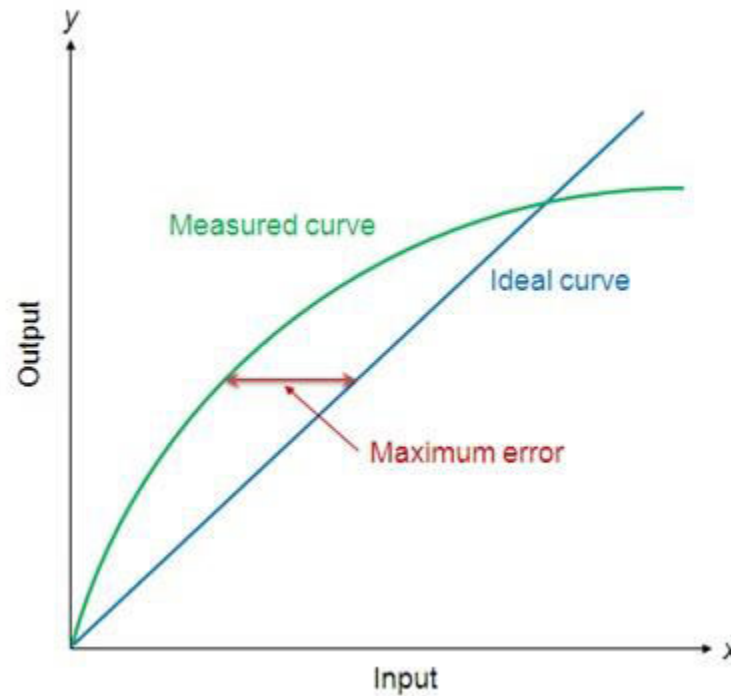
1. Range The range of a sensor indicates the limits between which the input can vary. For example, a thermocouple for the measurement of temperature might have a range of 25-225 °C.
2. Span- The span is difference between the maximum and minimum values of the input. Thus, the above-mentioned thermocouple will have a span of 200 °C.
3. Error- Error is the difference between the result of the measurement and the true value of the quantity being measured. A sensor might give a displacement reading of 29.8 mm, when the actual displacement had been 30 mm, then the error is -0.2 mm.

4. Accuracy- The accuracy defines the closeness of the agreement between the actual measurement result and a true value of the measurand. It is often expressed as a percentage of the full range output or full-scale deflection.

Example –A piezoelectric transducer used to evaluate dynamic pressure phenomena associated with explosions, pulsations, or dynamic pressure conditions in motors, rocket engines, compressors, and other pressurized devices is capable to detect pressures between 0.1 and 10,000 psig (0.7 KPa to 70 MPa). If it is specified with the accuracy of about $\pm 1\%$ full scale, then the reading given can be expected to be within ± 0.7 MPa.

5. Sensitivity -Sensitivity of a sensor is defined as the ratio of change in output value of a sensor to the per unit change in input value that causes the output change. For example, a general purpose thermocouple may have a sensitivity of $41 \mu\text{V}/^\circ\text{C}$.

6. Nonlinearity



The nonlinearity indicates the maximum deviation of the actual measured curve of a sensor from the ideal curve. Figure 2.1.1 shows a somewhat exaggerated relationship between the ideal, or least squares fit, line and the actual measured or *calibration* line. Linearity is often specified in terms of *percentage of nonlinearity*, which is defined as:

- Nonlinearity (%) = Maximum deviation in input / Maximum full scale input

7. Hysteresis - Hysteresis error curve The hysteresis is an error of a sensor, which is defined as the maximum difference in output at any measurement value within the sensor's specified range when approaching the point first with increasing and then with decreasing the input parameter.

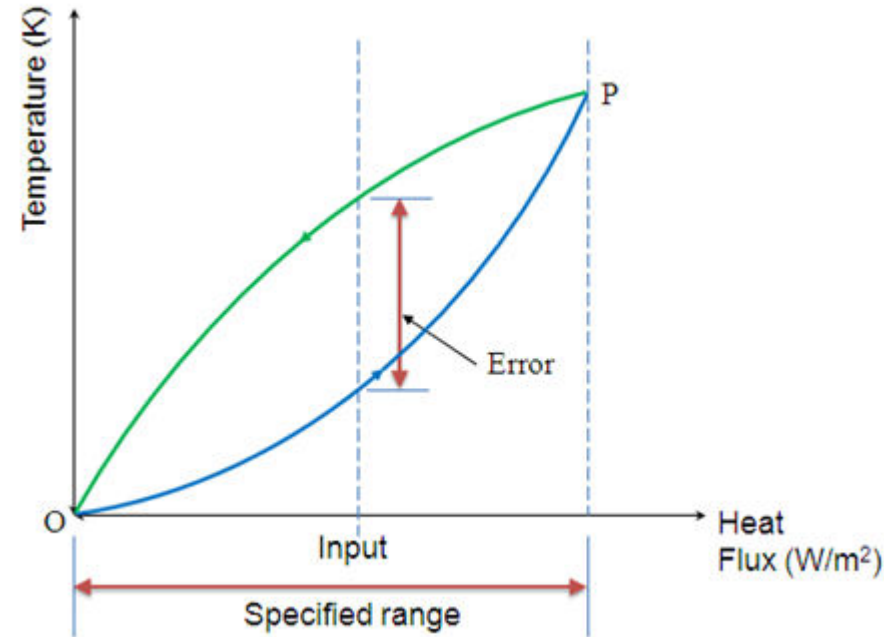


Figure above shows the hysteresis error might have occurred during measurement of using a thermocouple. The hysteresis error value is normally specified as a positive or negative percentage of the specified input range.

8. Resolution -Resolution is the smallest detectable incremental change of input parameter that can be detected in the output signal. Resolution can be expressed either as a proportion of the full-scale reading or in absolute terms. For example, if a LVDT sensor measures a displacement up to 20 mm and it provides an output as a number between 1 and 100 then the resolution of the sensor device is 0.2 mm.

9. Stability -Stability is the ability of a sensor device to give same output when used to measure a constant input over a period of time. The term 'drift' is used to indicate the change in output that occurs over a period of time. It is expressed as the percentage of full range output.

10. Dead band/time The dead band or dead space of a transducer is the range of input values for which there is no output. The dead time of a sensor device is the time duration from the application of an input until the output begins to respond or change

11. Repeatability- It specifies the ability of a sensor to give same output for repeated applications of same input value. It is usually expressed as a percentage of the full range output: $\text{Repeatability} = (\text{maximum} - \text{minimum values given}) \times 100 / \text{full range}$ (2)

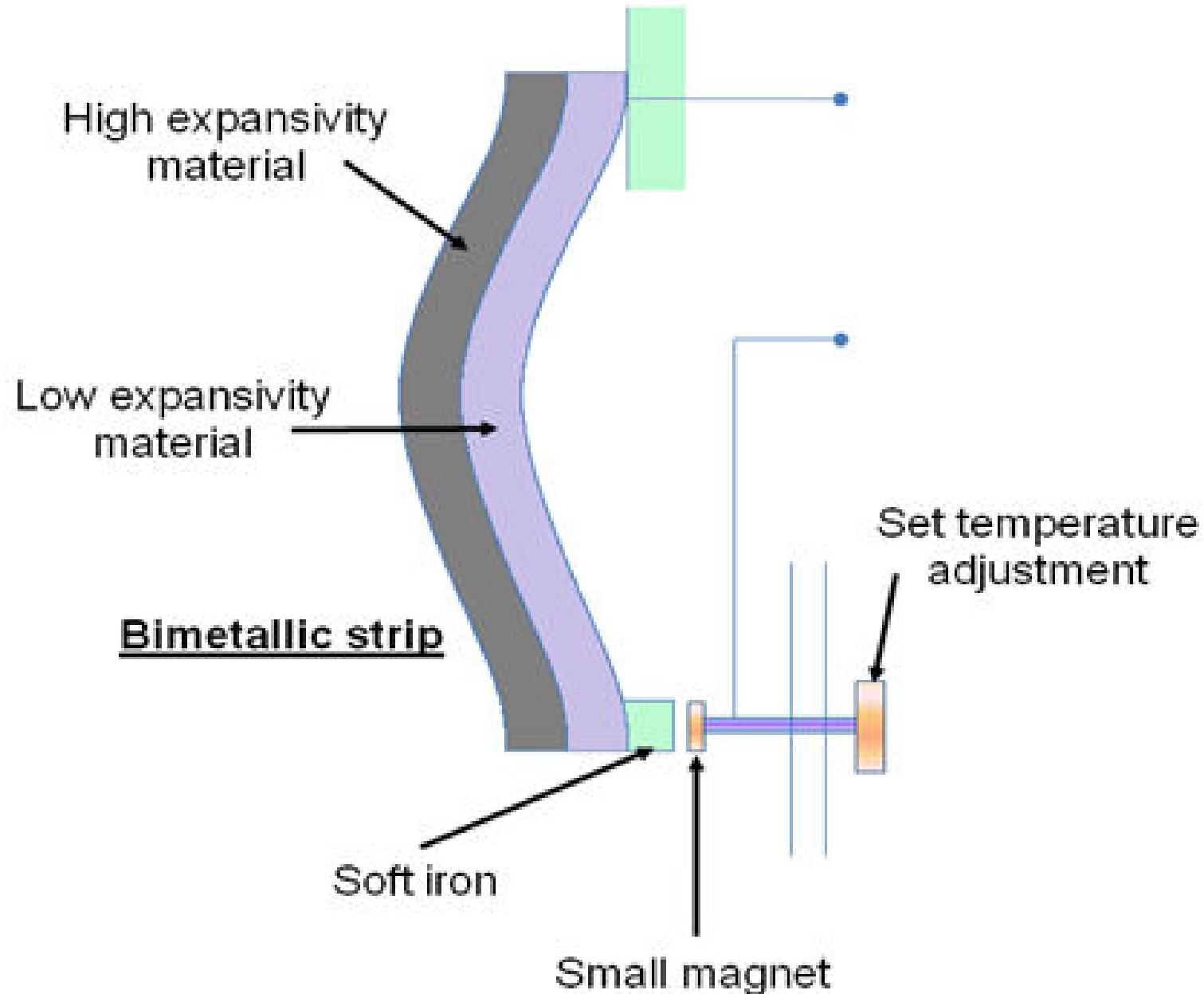
12. Response time- Response time describes the speed of change in the output on a step-wise change of the measurand. It is always specified with an indication of input step and the output range for which the response time is defined.

- Thermal sensor
- Optical
- pneumatic
- Electrical
- Acoustic
- Magnetic
- Piezoelectric sensors

Thermal sensors

- Temperature conveys the state of a mechanical system in terms of expansion or contraction of solids, liquids or gases, change in electrical resistance of conductors, semiconductors and thermoelectric emfs.
- Temperature sensors such as bimetallic strips, thermocouples, thermistors are widely used in monitoring of manufacturing processes such as casting, molding, metal cutting etc.
- The construction details and principle of working of some of the temperature sensors are discussed in following sections.

1. Bimetallic strips



Construction and working of Bi-metallic strip

- Bimetallic strips are used as thermal switch in controlling the temperature or heat in a manufacturing process or system.
- It contains two different metal strips bonded together.
- The metals have different coefficients of expansion.
- On heating the strips bend into curved strips with the metal with higher coefficient of expansion on the outside of the curve.
- Figure 2.5.1 shows a typical arrangement of a bimetallic strip used with a setting-up magnet.
- As the strips bend, the soft iron comes in closer proximity of the small magnet and further touches.
- Then the electric circuit completes and generates an alarm.
- In this way bimetallic strips help to protect the desired application from heating above the pre-set value of temperature.

1) Resistance Temperature Detector (RTD)

Detectors of wire resistance temperature common employ platinum, nickel or resistance wire elements, whose resistance variation with temperature has high intrinsic accuracy. They are available in many configurations and size and as shielded or open units for both immersion and surface applications.

The relationship between temperature and resistance of conductors can be calculated from the equation:

$$R = R_0 (1 + \alpha \Delta T)$$

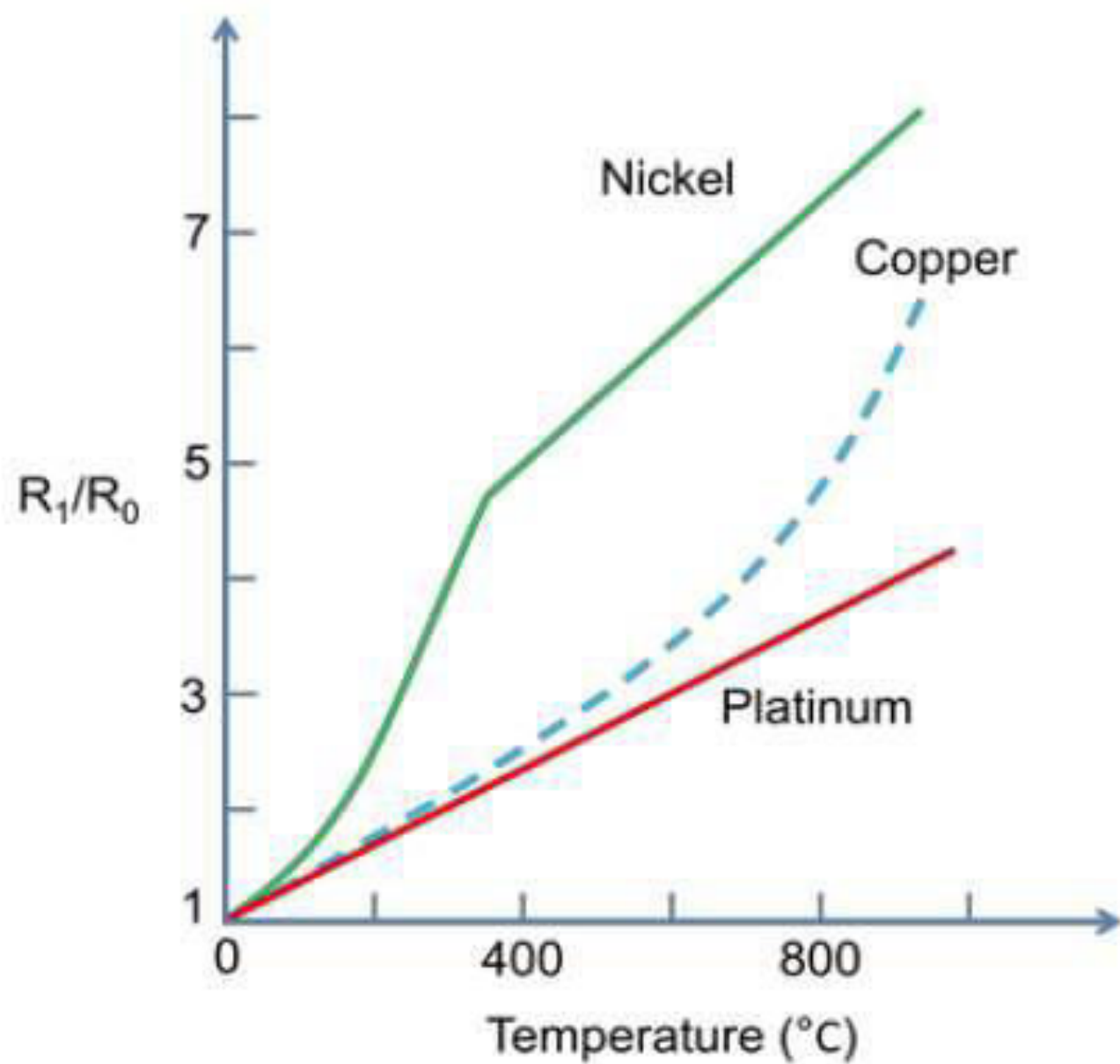
where

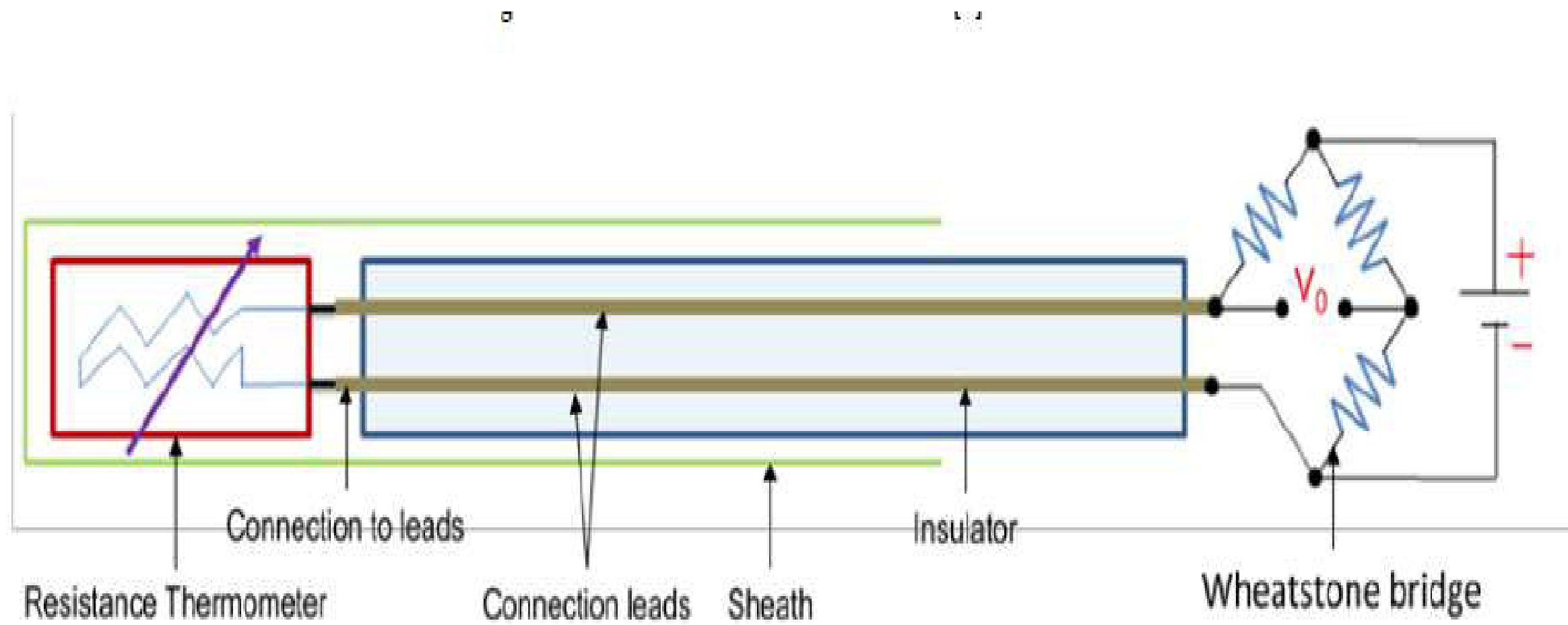
- R = the resistance of the conductor at temperature t ($^{\circ}\text{C}$)
- R_0 = the resistance at the reference temperature, usually 20°C
- α = the temperature coefficient of resistance
- ΔT = the difference between the operating and the reference temperature

1. Resistance temperature detectors (RTDs)

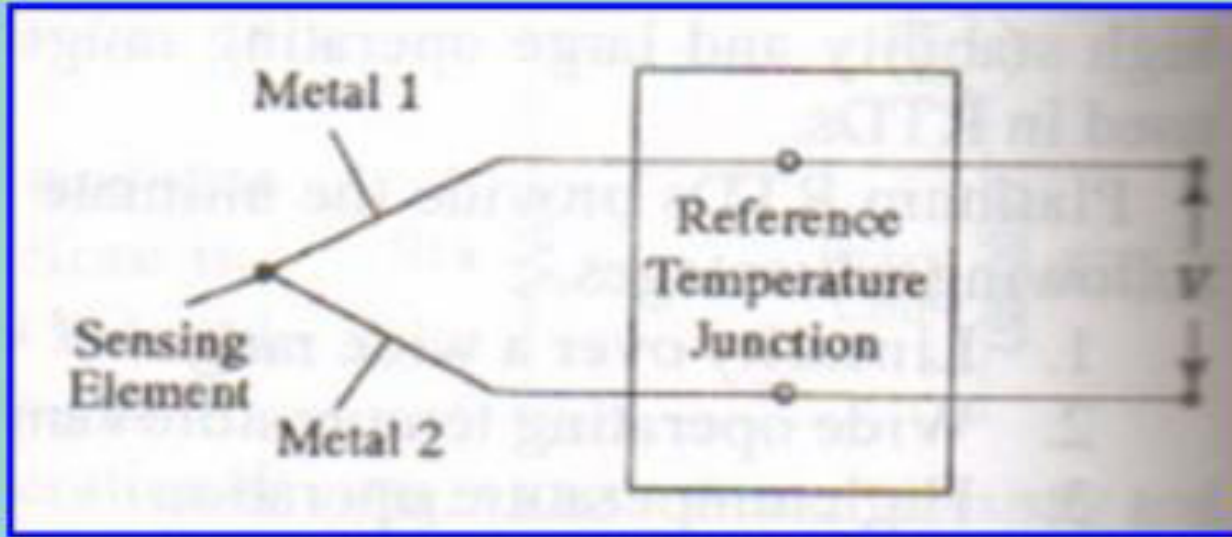
- RTDs work on the principle that the electric resistance of a metal changes due to change in its temperature.
- On heating up metals, their resistance increases and follows a linear relationship as shown in Figure 2.5.2.
- The correlation is $R_t = R_0 (1 + \alpha T)$ (2.5.1)

where R_t is the resistance at temperature T ($^{\circ}\text{C}$) and R_0 is the resistance at 0°C and α is the constant for the metal termed as temperature coefficient of resistance. The sensor is usually made to have a resistance of $100\ \Omega$ at 0°C





2) Thermocouple



It consists of two wires of different metals are joined together at one end, a temperature difference between this end and the other end of wires produces a voltage between the wires. The magnitude of this voltage depends on the materials used for the wires and the amount of temperature difference between the joined ends and the other ends.

The emf of the thermocouple :

$$E = c(T_1 - T_2) + k(T_1^2 - T_2^2)$$

Where

c and k = constant of the thermocouple materials

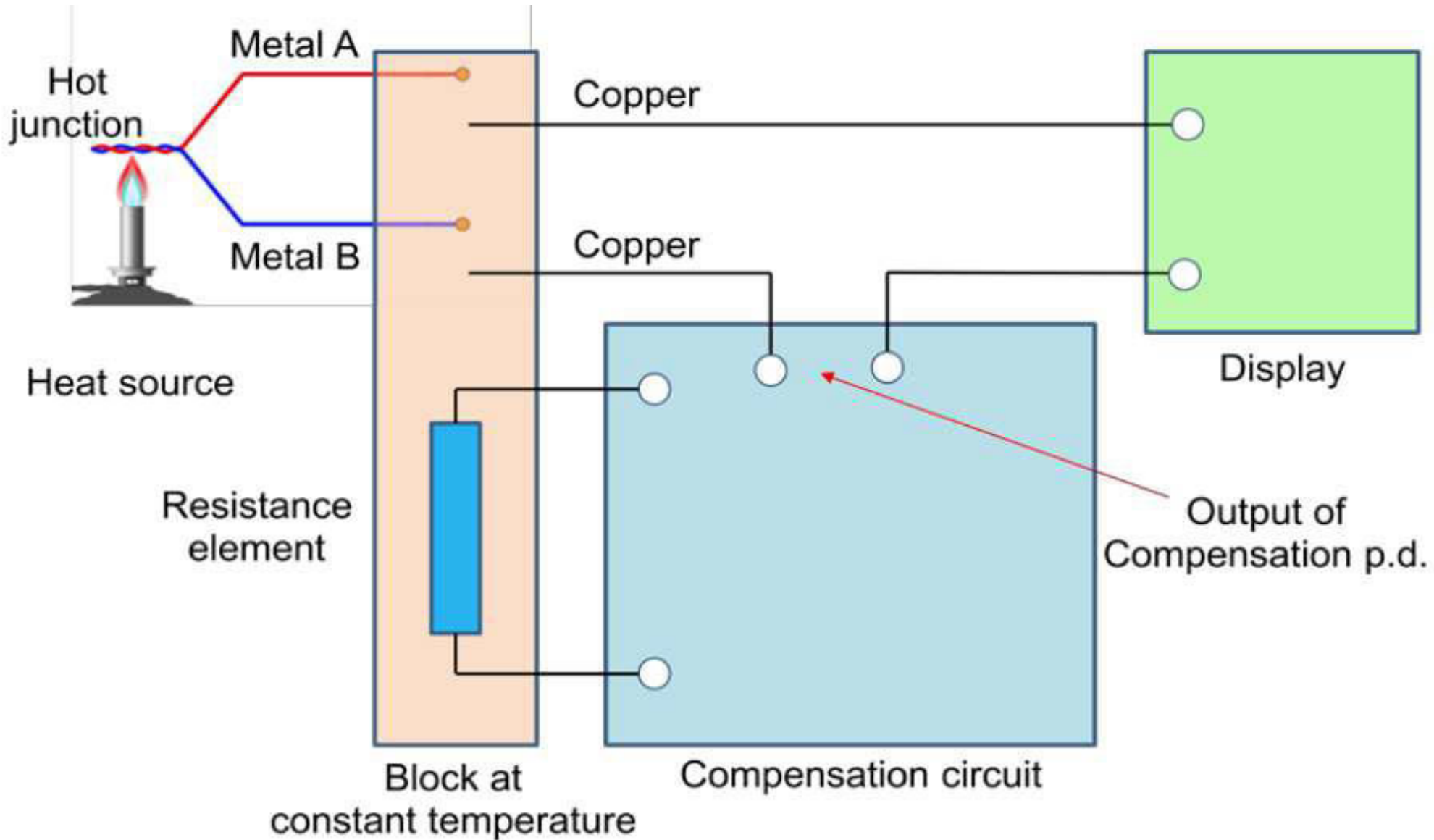
T_1 = The temperature of the “hot” junction

T_2 = The temperature of the “cold” or “reference” junction

- *Thermocouple working principle*

- Thermocouple works on the fact that when a junction of dissimilar metals heated, it produces an electric potential related to temperature.
- As per Thomas Seebeck (1821), when two wires composed of dissimilar metals are joined at both ends and one of the ends is heated, then there is a continuous current which flows in the thermoelectric circuit.
- Figure shows the schematic of thermocouple circuit. The net open circuit voltage (the Seebeck voltage) is a function of junction temperature and composition of two metals.
- It is given by, $\Delta V_{AB} = \alpha \Delta T$
where α , the Seebeck coefficient, is the constant of proportionality.
- Generally, Chromel (90% nickel and 10% chromium)–Alumel (95% nickel, 2% manganese, 2% aluminium and 1% silicon) are used in the manufacture of a thermocouple. Table shows the various other materials, their combinations and application temperature ranges.

Schematic of thermocouple circuit



Thermo couple material and Temperature range

Materials	Range (°C)	($\mu\text{V}/^\circ\text{C}$)
Platinum 30% rhodium/platinum 6% rhodium	0 to 1800	3
Chromel/constantan	-200 to 1000	63
Iron/constantan	-200 to 900	53
Chromel/alumel	-200 to 1300	41
Nirosil/nisil	-200 to 1300	28
Platinum/platinum 13% rhodium	0 to 1400	6
Platinum/platinum 10% rhodium	0 to 1400	6
Copper/constantan	-200 to 400	43

Application

- To monitor temperatures and chemistry throughout the steel making Process
- Testing temperatures associated with process plants e.g. chemical production and petroleum refineries
- Testing of heating appliance safety
- Temperature profiling in ovens, furnaces and kilns
- Temperature measurement of gas turbine and engine exhausts
- Monitoring of temperatures throughout the production and smelting process in the steel, iron and aluminium industry

3) Thermistor

A thermistor is a semiconductor made by sintering mixtures of metallic oxide, such as oxides of manganese, nickel, cobalt, copper and uranium.

Thermistors have negative temperature coefficient (NTC). That is, their resistance decreases as their temperature rises.

Types of thermistor

Disc

Resistance

1 to 1M Ω

Washer

1 to 50k Ω

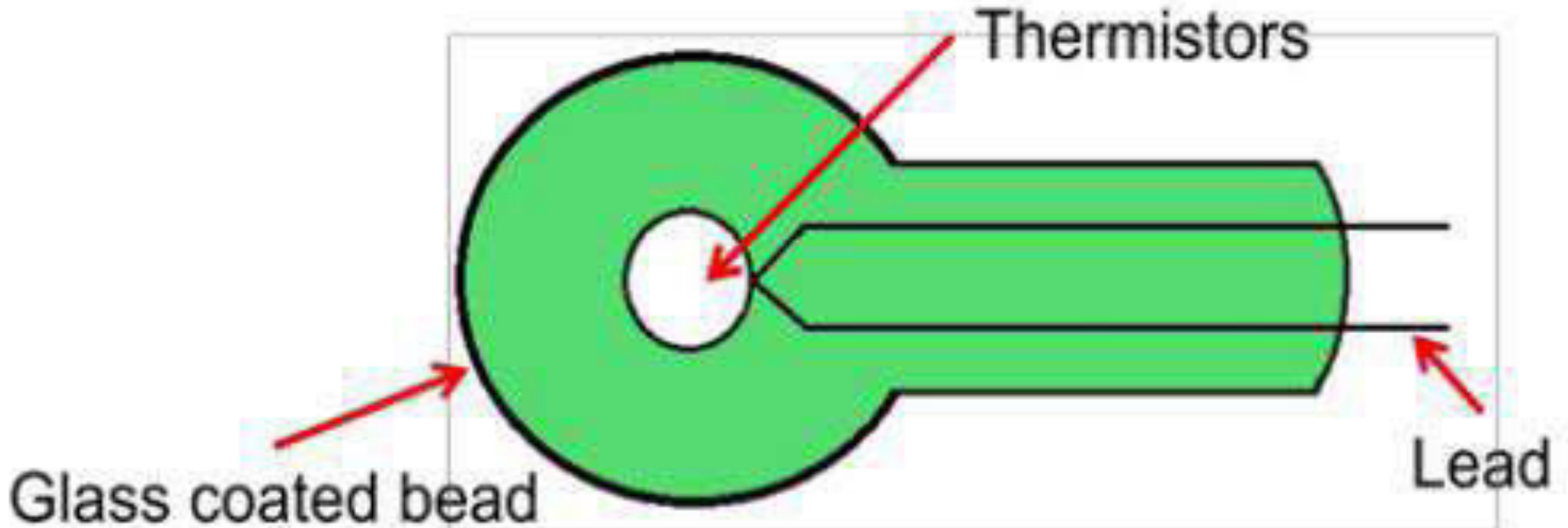
Rod

high resistance

---Thermistors-Principle

- Thermistors follow the principle of decrease in resistance with increasing temperature. The material used in thermistor is generally a semiconductor material such as a sintered metal oxide (mixtures of metal oxides, chromium, cobalt, iron, manganese and nickel) or doped polycrystalline ceramic containing barium titanate (BaTiO_3) and other compounds.
- As the temperature of semiconductor material increases the number of electrons able to move about increases which results in more current in the material and reduced resistance.
- Thermistors are rugged and small in dimensions. They exhibit nonlinear response characteristics.
- Thermistors are available in the form of a bead (pressed disc), probe or chip. Figure shows the construction of a bead type thermistor. It has a small bead of dimension from 0.5 mm to 5 mm coated with ceramic or glass material. The bead is connected to an electric circuit through two leads. To protect from the environment, the leads are contained in a stainless steel tube.

Schematic of a thermistor



Applications of Thermistors

- To monitor the coolant temperature and/or oil temperature inside the engine
- To monitor the temperature of an incubator
- Thermistors are used in modern digital thermostats
- To monitor the temperature of battery packs while charging
- To monitor temperature of hot ends of 3D printers
- To maintain correct temperature in the food handling and processing industry equipment's
- To control the operations of consumer appliances such as toasters, coffee makers, refrigerators, freezers, hair dryers, etc.

Advantages of thermistor

- Small size and low cost
- Fast response over narrow temperature range
- Good sensitivity in Negative Temperature Coefficient (NTC) region
- Cold junction compensation not required due to dependence of resistance on absolute temperature.
- Contact and lead resistance problems not encountered due to large resistance

Limitations of thermistor

- Non linearity in resistance vs temperature characteristics
- Unsuitable for wide temperature range
- Very low excitation current to avoid self heating
- Need of shielded power lines, filters, etc due to high resistance

Light sensors

- A light sensor is a device that is used to detect light.
- There are different types of light sensors such as photocell/photo resistor and photo diodes being used in manufacturing and other industrial applications.

2.1PHOTO RESISTOR

- Photo resistor is also called as light dependent resistor (LDR).
- It has a resistor whose resistance decreases with increasing incident light intensity.
- It is made of a high resistance semiconductor material, cadmium sulphide (CdS).
- The resistance of a CdS photo resistor varies inversely to the amount of light incident upon it.
- Photo resistor follows the principle of photoconductivity which results from the generation of mobile carriers when photons are absorbed by the semiconductor material.

construction of a photo resistor

- The CdS resistor coil is mounted on a ceramic substrate.
- This assembly is encapsulated by a resin material.
- The sensitive coil electrodes are connected to the control system through lead wires.
- On incidence of high intensity light on the electrodes, the resistance of resistor coil decreases which will be used further to generate the appropriate signal by the microprocessor via lead wires.

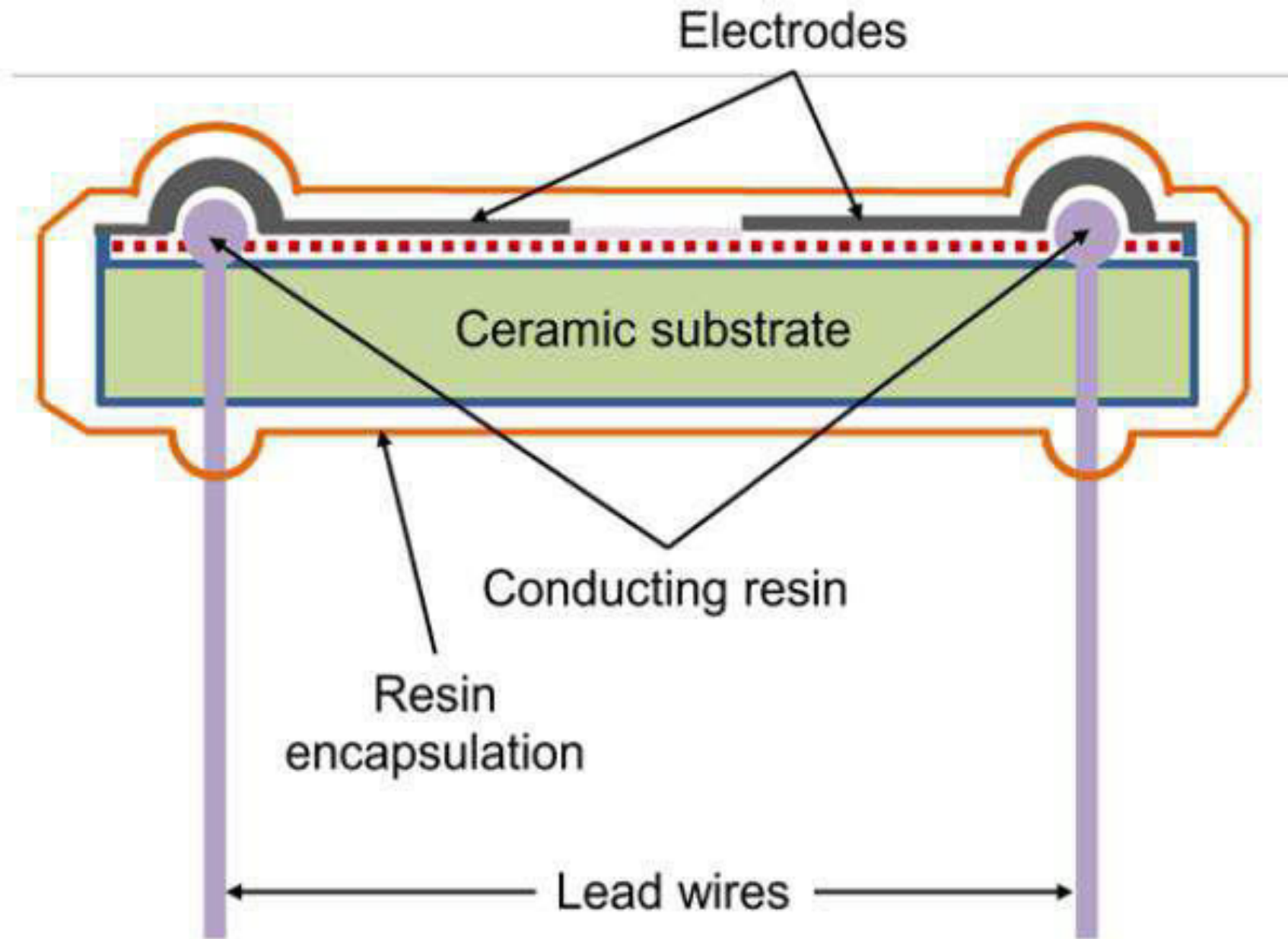


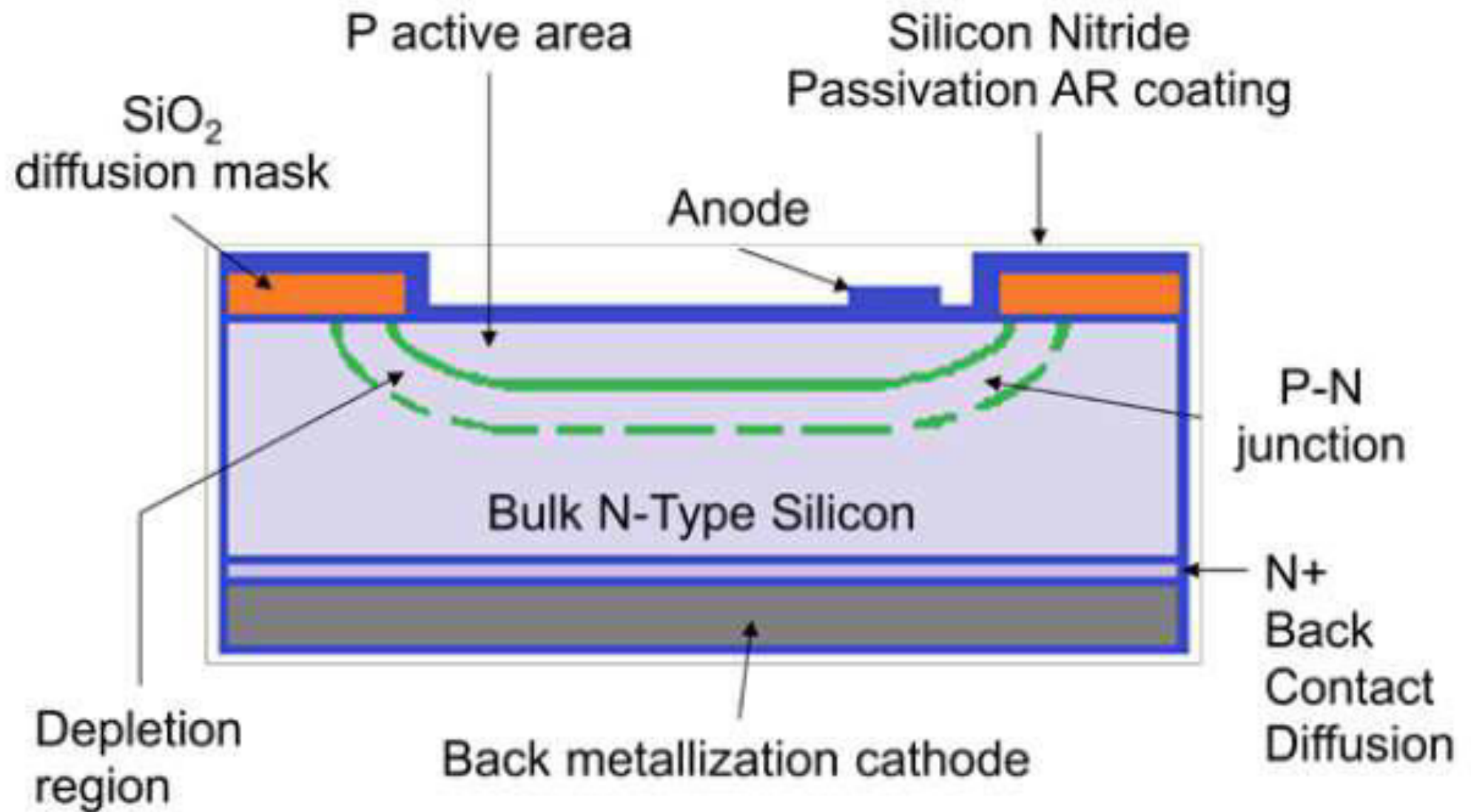
Photo resistors are used in science and in almost any branch of industry for control, safety, amusement, sound reproduction, inspection and measurement.

Applications of photo resistor

- Computers, wireless phones, and televisions, use ambient light sensors to automatically control the brightness of a screen
- Barcode scanners used in retailer locations work using light sensor technology
- In space and robotics: for controlled and guided motions of vehicles and robots. The light sensor enables a robot to detect light. Robots can be programmed to have a specific reaction if a certain amount of light is detected.
- Auto Flash for camera
- Industrial process control

2.Photo diodes

- Photodiode is a solid-state device which converts incident light into an electric current. It is made of Silicon. It consists of a shallow diffused p-n junction, normally a p-on-n configuration. When photons of energy greater than 1.1eV (the band gap of silicon) fall on the device, they are absorbed and electron hole pairs are created.
- The depth at which the photons are absorbed depends upon their energy. The lower the energy of the photons, the deeper they are absorbed. Then the electron-hole pairs drift apart. When the minority carriers reach the junction, they are swept across by the electric field and an electric current establishes
- Photodiodes are one of the types of photo detector, which convert light into either current or voltage. These are regular semiconductor diodes except that they may be either exposed to detect vacuum UV or X-rays or packaged with an opening or optical fiber connection to allow light to reach the sensitive part of the device.



Photodiode

construction of Photo diode detector

- Figure shows the construction of Photo diode detector.
- It is constructed from single crystal silicon wafers.
- It is a p-n junction device.
- The upper layer is p layer. It is very thin and formed by thermal diffusion or ion implantation of doping material such as boron.
- Depletion region is narrow and is sandwiched between p layer and bulk n type layer of silicon.
- Light irradiates at front surface, anode, while the back surface is cathode.
- The incidence of light on anode generates a flow of electron across the p-n junction which is the measure of light intensity.

Applications of photo diodes

- **Camera:** Light Meters, Automatic Shutter Control, Auto-focus, Photographic Flash Control
- **Medical:** CAT Scanners - X ray Detection, Pulse Oximeters, Blood Particle Analyzers

Industry

- Bar Code Scanners
- Light Pens
- Brightness Controls
- Encoders
- Position Sensors
- Surveying Instruments
- Copiers - Density of Toner

Safety Equipment

- Smoke Detectors
- Flame Monitors
- Security Inspection Equipment - Airport X ray
- Intruder Alert - Security System

Automotive

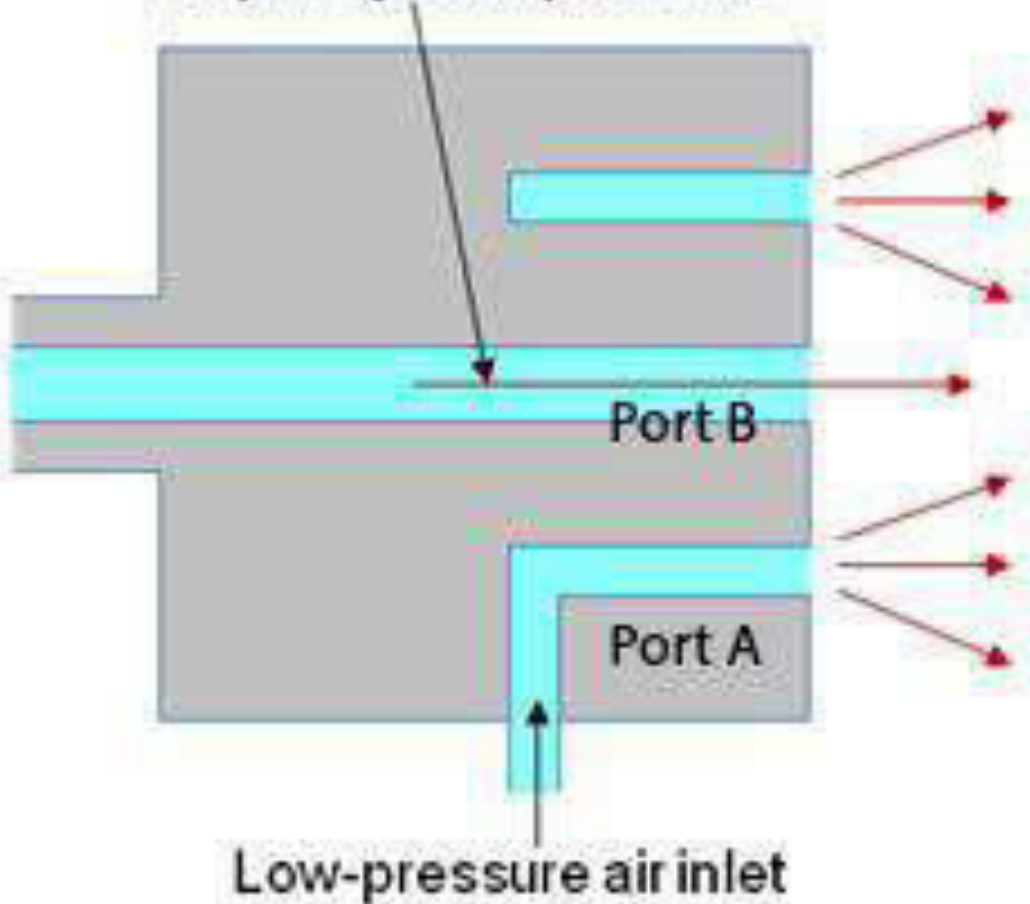
- Headlight Dimmer
- Twilight Detectors
- Climate Control - Sunlight Detector

Communications

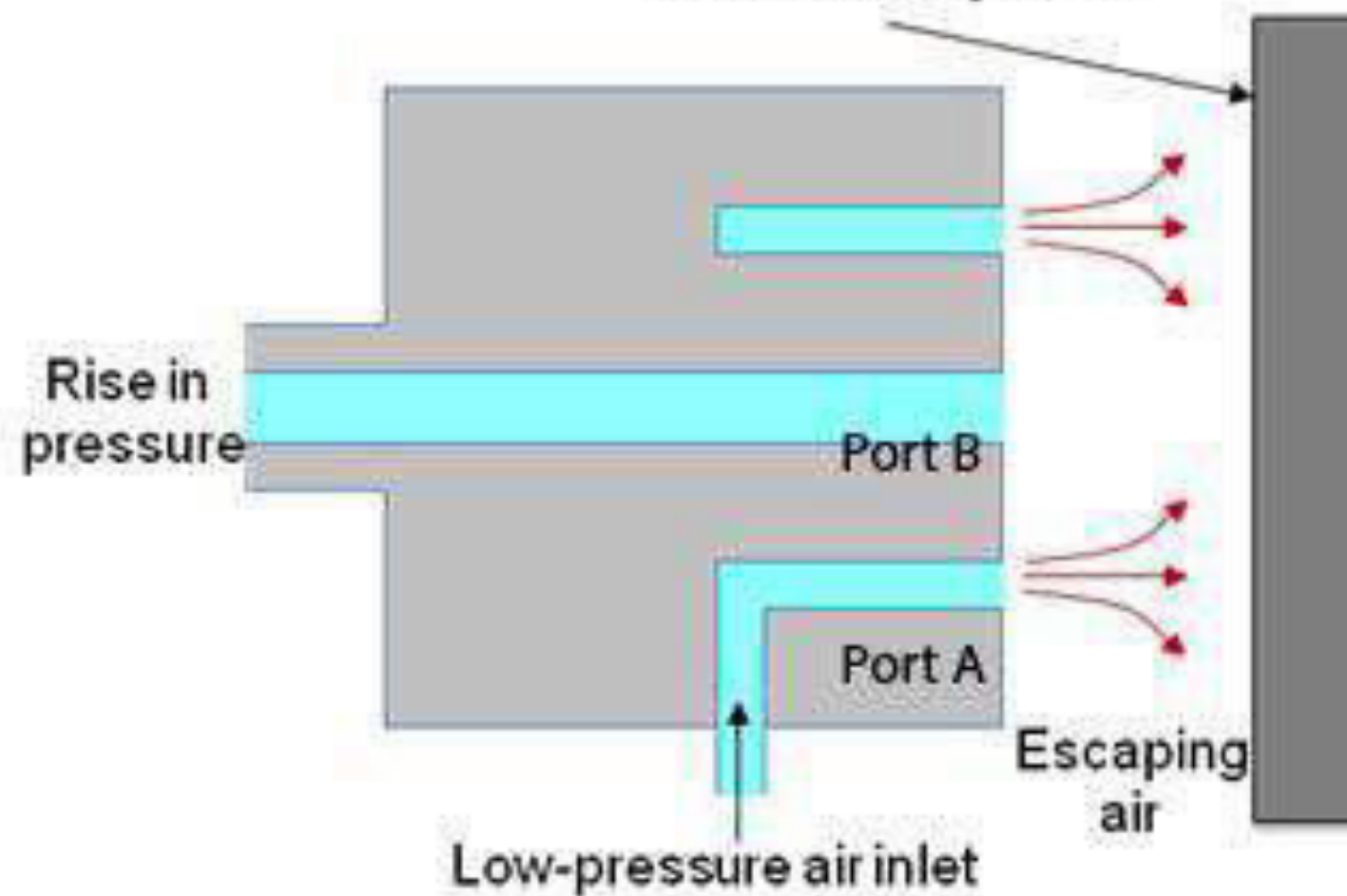
- Fiber Optic Links
- Optical Communications
- Optical Remote Control

Pneumatic Sensors

Air dragged out of port and so
Drop in system pressure



Object blocking escaping air increases
Pressure in system



Working of Pneumatic Sensors

- Pneumatic sensors are used to measure the displacement as well as to sense the proximity of an object close to it.
- The displacement and proximity are transformed into change in air pressure.
- Figure shows a schematic of construction and working of such a sensor. It comprises of three ports. Low pressure air is allowed to escape through port A. In the absence of any obstacle / object, this low pressure air escapes and in doing so, reduces the pressure in the port B. However when an object obstructs the low pressure air (Port A), there is rise in pressure in output port B. This rise in pressure is calibrated to measure the displacement or to trigger a switch. These sensors are used in robotics, pneumatics and for tooling in CNC machine tools.